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Clinical Features and Natural Progression of Unilateral High Myopia in Adults: A Comparison Study

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Purpose: To investigate and compare the clinical characteristics of patients with unilateral high myopia (UHM) and bilateral high myopia (BHM) based on axial length (AL).

Design: A retrospective cohort study.

Participants: Adult patients diagnosed with UHM or BHM between March 2011 and August 2021.

Methods: Unilateral high myopia was defined as ≥26 mm AL in 1 eye and <26 mm in the other, with ≥2 mm difference. Bilateral high myopia was defined as ≥26 mm AL in both eyes, with ≤3 mm difference. In each patient, the eye with the longer AL was designated the "longer eye" and the other the "shorter eye." We analyzed differences in clinical features, including ophthalmic history, best-corrected visual acuity, ocular biometry, and myopic maculopathy grade. Myopic maculopathy was graded based on atrophy, traction, and neovascularization using a known method. Long-term features included treatments for myopic neovascular maculopathy and myopic tractional maculopathy and AL change over time.

Main Outcome Measures: Comparison of clinical characteristics between UHM and BHM groups.

Results: We analyzed 369 patients (79 with UHM and 290 with BHM) with a median follow-up period of 4.5 years. The UHM group had a higher proportion of women than the BHM group (88.8% vs. 76.2%, $P = 0.025$). Compared with longer eyes in the BHM group, those in the UHM group had worse best-corrected visual acuity (0.8 ± 0.6 vs. 0.6 ± 0.6 in logarithm of the minimum angle of resolution, $P < 0.001$) despite having shorter AL (29.1 ± 1.6 mm vs. 30.6 ± 1.9 mm, $P < 0.001$). In the analysis of AL changes, shorter eyes in the UHM group showed no elongation over time (0.014 mm/year, $P = 0.12$), unlike the longer eyes in UHM and both eyes in BHM (0.049–0.071 mm/year, $P < 0.01$).

Conclusions: Adult UHM patients mostly lacked associated environmental factors. The poorer visual acuity in the longer eyes of UHM patients, which cannot be explained by structural abnormalities, suggests that the interocular difference may have originated in early childhood. During the follow-up period, AL elongation and myopic complications occurred at similar rates in the longer eye of UHM and both eyes of BHM. Meanwhile, such changes were not observed in the shorter eye in UHM. Further investigation of the underlying mechanisms, such as the genetic factors contributing to this extreme asymmetry, is warranted.

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The prevalence of myopia, the most common ocular disease, is on the rise.[1,2] A meta-analysis, which synthesized studies conducted between 1995 and 2015, reported that approximately 22.9% (1.4 billion people) of the global population had myopia.[3] High myopia, defined by a spherical equivalent of less than −5 or −6 diopters and an axial length (AL) of >26.0 or 26.5 mm, is estimated to be prevalent in 0.9% to 3.1% of the global population.[4,5] However, in regions like East Asia, where myopia rates are notably high, its prevalence ranges from 6.8% to 21.6%.[6,7] Projections indicate that by 2050, around 49.8% (approximately 4.7 billion people) of the global population will have myopia, with 9.8% (938 million people) having high myopia.[3]

Myopia is a multifaceted, multifactorial condition influenced by environmental factors such as hyperopic defocus, exposure to bright light, and vitamin D or dopamine.[8–10] It can also be inherited, with several genetic loci associated with the conditions identified.[11,12] However, the mechanisms through which these factors affect myopia and the extent of their impact remain unknown.[13] Furthermore, in a subset of patients, the condition cannot be explained simply through genetic or environmental factors, such as those with unilateral high myopia (UHM), exhibiting distinct AL and fundus findings in each eyes. Although further research on various factors is warranted, it is worth noting that some individuals exhibit high myopia in only 1 eye, even in adulthood, despite sharing similar genetic and

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environmental influences between both eyes. Identifying the underlying cause in these cases may provide new perspectives on the development of myopia. However, the clinical features of these patients are inadequately documented, with only limited reported case series.[14,15] Additionally, there is a lack of studies on how UHM differs in clinical features from that of general high myopia. Therefore, this study aimed to compare the clinical features of patients with UHM to those of patients with bilateral high myopia (BHM), diagnosed based on ALs.

Methods

This retrospective cohort study was conducted at the Seoul National University Bundang Hospital, Republic of Korea, and involved patients from the Bundang High Myopia Cohort. The study adhered to the tenets of the Declaration of Helsinki and was approved by the Institutional Review Board of Seoul National University Bundang Hospital (Institutional Review Board approval number: B-2203-744-108). Owing to the nature of this study, written informed consent was waived. The inclusion criteria were as follows: (1) Patients with an AL of $\geq$26 mm in 1 eye and $<$26 mm in the other, with a difference of $\geq$2 mm in AL between the eyes, classified as the UHM group; (2) Patients with an AL of $\geq$26 mm in both eyes and a difference of $\leq$3 mm between the eyes, classified as the BHM group. To exclude patients in the gray zone, we included only cases with an interocular AL difference of $\leq$3 mm in the BHM group. For subgroup comparisons, we further divided the groups into longer eyes (eyes with greater AL) and shorter eyes (eyes with smaller AL).

A comprehensive ophthalmic examination, including best-corrected visual acuity, lens status, refractive error, color fundus photography, OCT (Spectralis; Heidelberg Engineering), and ocular biometry (Carl Zeiss Meditec AG), was conducted at the initial visit. We collected data from all examinations and treatments administered during the follow-up period. Myopia-related treatments were compared for 2 specific targets: myopic neovascular maculopathy (MNM) and myopic traction maculopathy (MTM).

We conducted survival analysis using the Kaplan–Meier method to estimate the cumulative probability of the first occurrence of MNM or MTM in treatment-naïve eyes. Log-rank tests were employed to compare survival curves between the UHM and BHM groups. Cox proportional hazards models were used to adjust for potential confounding factors, such as age and baseline AL. Statistical significance was set at a P value of $<$0.05. Treatment-naïve patients were defined as follows: for MNM, patients who had not received any prior intravitreal injections (e.g., anti-VEGF agents) or macular laser treatment, including photodynamic therapy; for MTM, patients who had not undergone pars plana vitrectomy or macular buckling surgery.

Myopic maculopathy was graded according to the method proposed by Ruiz-Medrano et al.[16] Using this method, we classified myopic maculopathy into 3 components: atrophy, traction, and neovascularization. We adopted a modified method in which N1 (macular lacquer cracks), N2a (active choroidal neovascularization [CNV]), and N2s (scar/Fuch spot) were collapsed to N1 (myopic CNV) for simplicity due to the complexity of grading such detailed neovascular components using our data. Two retinal specialists (D.G.K. and D.J.K.) independently conducted the grading, and intergrader agreements were measured. In cases where an agreement cannot be reached, a senior retinal specialist (K.H.P.) adjudicated all discrepancies. We excluded patients without initial fundus photography and OCT or those with severe macular comorbidities that rendered myopic maculopathy grading impossible. Additionally, 2 retinal specialists (S.H.B. and D.G.K.) analyzed other OCT parameters, such as central macular thickness, subfoveal choroidal thickness, and dome-shaped macula.

Statistical Analysis

For statistical analyses, continuous data were presented as mean $\pm$ standard deviation, while categorical variables were presented as number and percentage. Continuous variables were first assessed for normality using the Shapiro–Wilk test. Variables with normal distributions were analyzed using Welch t test to account for unequal variances, while variables that did not meet the normality assumption were analyzed using the Mann–Whitney U test. Categorical variables were analyzed using Pearson chi-square test or Fisher exact test, depending on the expected frequencies. To assess the potential impact of confounders, logistic regression analysis was also performed when appropriate. The intergrader agreement of the myopic maculopathy grading was analyzed using Cohen weighted kappa. The generalized estimating equation (GEE) was used to analyze changes in AL over time. Each time point of the AL examination and the patient's surgical history were matched, and cataract surgery or pars plana vitrectomy performed between measurements were included as fixed effects in the statistical analysis. All statistical analyses were performed using SPSS (version 23.0; IBM Corp) for the GEE and R language version 4.1.2 (R Foundation for Statistical Computing) for other analyses. Statistical significance was set at a P value of $<$0.05. For post hoc pairwise comparisons of categorical variables with multiple subcategories, the Bonferroni test was used to adjust the alpha level, setting the significance threshold at 0.05 divided by the total number of pairwise comparisons.

Results

We analyzed 422 patients from the Bundang High Myopia Cohort from March 8, 2011, to August 5, 2021. Among them, 27 patients without available color fundus photography or OCT for both eyes at the initial visit and 26 patients with macular comorbidities (21 cases of rhegmatogenous retinal detachment, 3 cases of macular sequelae of previous rhegmatogenous retinal detachment, and 2 cases of branch retinal vein occlusion) that made the grading of myopic maculopathy impossible were excluded. Ultimately, we included 369 patients in our final analysis, with 79 and 290 patients categorized into the UHM and BHM groups, respectively. Figure 1 provides an overview of the included patients.

The baseline characteristics of the included patients are presented in Table 1. The mean age was 57.1 $\pm$ 17.3 years in the UHM group and 54.4 $\pm$ 12.7 years in the BHM group (P = 0.199). The proportion of females was higher in the UHM group than in the BHM group (70 patients [88.8%] vs. 221 patients [76.1%], P = 0.025). In the UHM group, both the longer eye (29.1 $\pm$ 1.6 vs. 30.6 $\pm$ 1.9 mm, P = 0.000) and the shorter eye (24.4 $\pm$ 1.0 vs. 29.7 $\pm$ 2.0 mm, P = 0.000) had shorter ALs compared with those in the BHM group. However, the intereye AL difference was significantly larger in the UHM group (4.7 $\pm$ 1.7 vs. 0.9 $\pm$ 0.8 mm, P = 0.000). The follow-up period of each group was not significantly different (4.8 $\pm$ 4.5 vs. 5.7 $\pm$ 4.6 years, P = 0.124).

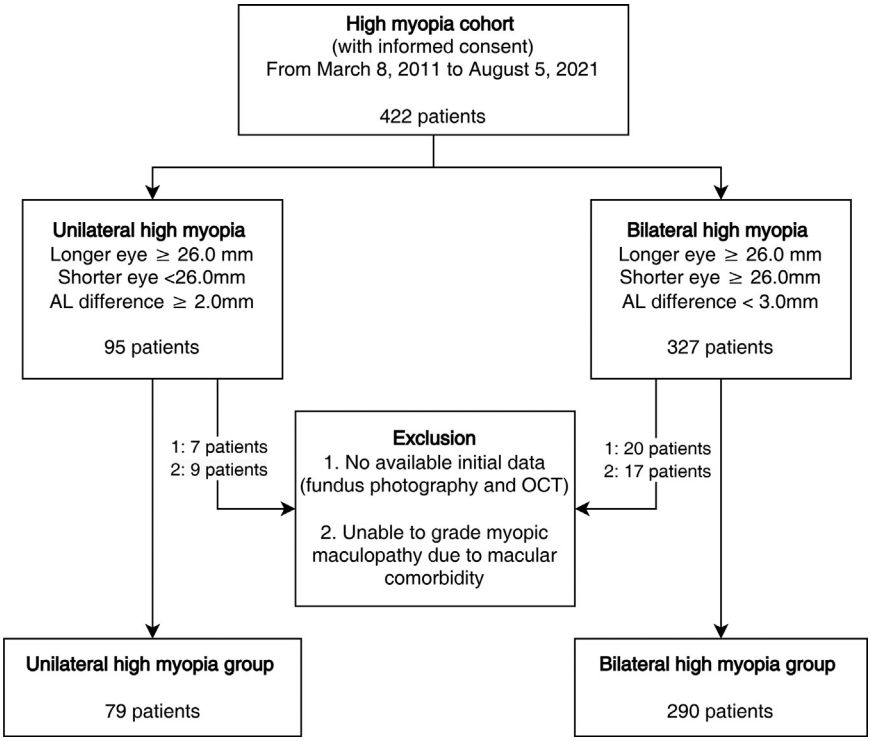


Figure 1. Flow chart of the study population. AL = axial length.

For longer eyes, the mean refractive error, lens status, previous refractive surgery, previous CNV treatment, and previous vitrectomy rates were similar between the groups. Despite the UHM group having a shorter AL, the longer eyes in this group had worse best-corrected visual acuity (0.8 ± 0.6 vs. 0.6 ± 0.6 in logarithm of the minimum angle of resolution, $P = 0.000$). When comparing shorter eyes, significant differences were observed between the 2 groups in all aspects, except for lens status ($P = 0.299$) and the proportion of previous vitrectomy ($P = 0.195$). Notably, there was no history of CNV treatment in shorter eyes in the UHM group, whereas, in the BHM group, 40 patients (13.8%) had previously undergone CNV treatment, a rate similar to those of longer eyes in both the UHM (10 patients, 12.7%) and BHM groups (41 patients, 14.1%). We also performed logistic regression analysis for potential confounders, but the significances of the variables that met the linear assumption were consistent with the above results.

While most patients with UHM did not have any congenital or acquired ophthalmic history that could cause significant asymmetry between the eyes, 1 patient underwent penetrating keratoplasty for keratoconus in the longer eye at 30 years, and 3 reported a decline in vision in their longer eyes after childhood trauma. Aside from 1 patient with band keratopathy, there were no notable findings except for myopic degeneration in these cases. Other patients in both groups showed no remarkable findings except for myopic changes in the longer eye. Notably, all patients who underwent cataract surgery at the initial examination did so after the age of 53 years.

Myopic maculopathy grade and other OCT parameters at the initial examination were compared between the 2 groups (Table 2). In the comparison of longer eyes, there were no significant differences between the 2 groups in the grades of any of the 3 components of myopic maculopathy. Additionally, there were no significant differences between the groups in terms of central macular thickness, subfoveal choroidal thickness, or proportion of dome-shaped maculae. In contrast, significant differences were observed between shorter eyes in both groups. The shorter eyes in the UHM group exhibited only 3 cases of diffuse chorioretinal atrophy, 1 case of inner or outer foveoschisis, and 1 case of a full-thickness macular hole, whereas the remaining patients did not show any pathological myopia features. Furthermore, the UHM group's shorter eyes showed a thicker subfoveal choroidal thickness (205.0 ± 88.6 vs. 92.0 ± 88.6, $P = 0.000$), and a lower proportion of dome-shaped macula (2.5% [2 patients] vs. 21.4% [62 patients], $P = 0.000$) compared with those in the BHM group. The intergrader agreements (kappa values) for atrophy, traction, and neovascularization were 0.94, 0.95, and 0.97, respectively.

We conducted survival analysis on treatment-naïve patients based on the timing of their first treatment during the follow-up period (Fig 2). For longer eyes, the cumulative probability of receiving the first treatment for MNM by 5 years was 11.7 ± 3.2% (BHM group) and 14.9 ± 5.7% (UHM group), showing no statistically significant difference in risk between the 2 groups ($P = 0.7$). For shorter eyes, the cumulative probability of receiving the first treatment for MNM by 5 years in the BHM group

Table 1. Patients' Baseline and Demographic Characteristics

Characteristics	UHM	BHM	P Value
Number of patients	79	290	
Age, yrs	57.1 ± 17.3	54.4 ± 12.7	0.091
Female, number (%)	70 (88.8%)	221 (76.2%)	0.025
Axial length, mm			
Longer eye (greater axial length)	29.1 ± 1.6	30.6 ± 1.9	0.000
Shorter eye (lesser axial length)	24.4 ± 1.0	29.7 ± 2.0	0.000
Difference	4.7 ± 1.7	0.9 ± 0.8	0.000
Follow-up period, yrs	4.8 ± 4.5	5.7 ± 4.6	0.108
Longer eye			
BCVA (logMAR)	0.8 ± 0.6	0.6 ± 0.6	0.000
Refractive error (diopters)	−9.0 ± 6.3	−10.7 ± 7.1	0.064
Previous refractive surgery	29 (36.7%)	112 (38.6%)	0.858
Lens status			0.388
Phakia	56 (70.9%)	226 (77.9%)	
Pseudophakia	21 (26.6%)	60 (20.7%)	
Aphakia	2 (2.5%)	4 (1.4%)	
Previous CNV treatment	10 (12.7%)	41 (14.1%)	0.878
Previous vitrectomy	1 (1.3%)	9 (3.1%)	0.616
Shorter eye			
BCVA (logMAR)	0.1 ± 0.2	0.5 ± 0.5	0.000
Refractive error (diopters)	−1.7 ± 2.2	−9.5 ± 6.5	0.000
Previous refractive surgery	17 (21.5%)	113 (39.0%)	0.006
Lens status			0.299
Phakia	67 (84.8%)	225 (77.6%)	
Pseudophakia	12 (15.2%).	62 (21.4%)	
Aphakia	0	3 (1.0%)	
Previous CNV treatment	0	40 (13.8%)	0.001
Previous vitrectomy	1 (1.3%)	16 (5.5%)	0.195

BCVA = best-corrected visual acuity; BHM = bilateral high myopia; CNV = choroidal neovascularization; logMAR = logarithm of the minimum angle of resolution; UHM = unilateral high myopia.

was 6.6 ± 3.2%, whereas no MNM treatment occurred in the UHM group during the follow-up period ($P = 0.02$). For the treatment of MTM in the longer eyes, the cumulative probability by 5 years was 7.2% ± 1.8% for the BHM group and 8.8% ± 4.7% for the UHM group, indicating no significant difference between the groups ($P = 0.61$). In the shorter eyes, while the BHM group had a cumulative probability of 3.6% at 5 years for MTM treatment, the UHM group had no such occurrences during the study period ($P = 0.04$).

Adult patients who underwent at least 1 follow-up examination of AL (except for data measured repeatedly within 1 month) were analyzed (Table 3 and Fig 3). A total of 716 AL examinations were performed on 222 adult patients. The mean number of AL measurements per patient was 3.2 ± 1.3, and the mean follow-up period from baseline for each examination was 4.8 ± 3.4 years. All follow-up data on AL change are depicted as spaghetti plots in Figure 3. When analyzing factors influencing repeatedly measured AL using GEE, cataract surgery performed during the follow-up period resulted in a shorter subsequent AL compared with preoperative measurements (−0.182 ± 0.038 for the longer eye, $P < 0.01$; 0.156 ± 0.070 mm for the shorter eye, $P = 0.03$). Sex and pars plana vitrectomy during the follow-up period showed no statistically significant effect on AL in either eye. In the UHM group, the longer eye showed a statistically significant positive correlation with time, indicating an increase in AL over time (0.049 ± 0.010 mm/year, $P < 0.01$), but there was no significant correlation in the shorter eye (0.014 ± 0.008 mm/year, $P = 0.12$). In contrast, both eyes in the BHM group exhibited significant axial elongation over time (0.066 ± 0.006 mm/year for the longer eye,

Table 2. Comparison of Myopic Maculopathy Grade and Other OCT Parameters at the Initial Presentation

Characteristics	Longer Eye			Shorter Eye		
	UHM	*BHM*	P *Value*	*UHM*	*BHM*	P *Value*
Atrophy*			0.156			0.000
Normal or tessellated fundus	22 (27.8%)	47 (16.2%)		76 (96.2%)	85 (29.3%)	
Diffuse chorioretinal atrophy	29 (36.7%)	133 (45.9%)		3 (3.8%)	123 (42.4%)	
Patchy chorioretinal atrophy	20 (25.3%)	84 (29.0%)		0	63 (21.7%)	
Complete macular atrophy	8 (10.1%)	26 (9.0%)		0	19 (6.6%)	
Traction*			0.219			0.002
No macular schisis	50 (63.3%)	209 (72.1%)		77 (97.5%)	221 (76.6%)	
Inner or outer foveoschisis	20 (25.3%)	41 (14.1%)		1 (1.3%)	38 (13.1%)	
Inner and outer foveoschisis	8 (10.1%)	30 (10.3%)		0	13 (4.5%)	
Foveal detachment	0	3 (1.0%)		0	9 (3.1%)	
Full-thickness MH	1 (1.3%)	4 (1.4%)		1 (1.3%)	4 (1.4%)	
MH with retinal detachment	0	3 (1.0%)		0	4 (1.4%)	
Neovascularization*			1.00			0.000
No myopic CNV	65 (82.3%)	240 (82.8%)		79 (100.0%)	235 (81.0%)	
Myopic CNV	14 (17.7%)	50 (17.2%)		0	55 (19.0%)	
Other OCT parameters						
Central macular thickness, μm	259.8 ± 118.6	248.9 ± 100.3	0.457	248.0 + 56.1	273.1 ± 134.1	0.552
Subfoveal choroidal thickness, μm	74.3 ± 57.5	74.3 ± 55.3	0.866	205.0 ± 88.6	92.0 ± 88.6	0.000
Dome-shaped macula, n (%)	16 (20.3%)	93 (32.1%)	0.057	2 (2.5%)	62 (21.4%)	0.000

BHM = bilateral high myopia; CNV = choroidal neovascularization; MH = macular hole; UHM = unilateral high myopia.
*Intergrade agreement (Cohen weighted kappa): atrophy (0.94), traction (0.95), and neovascularization (0.97).

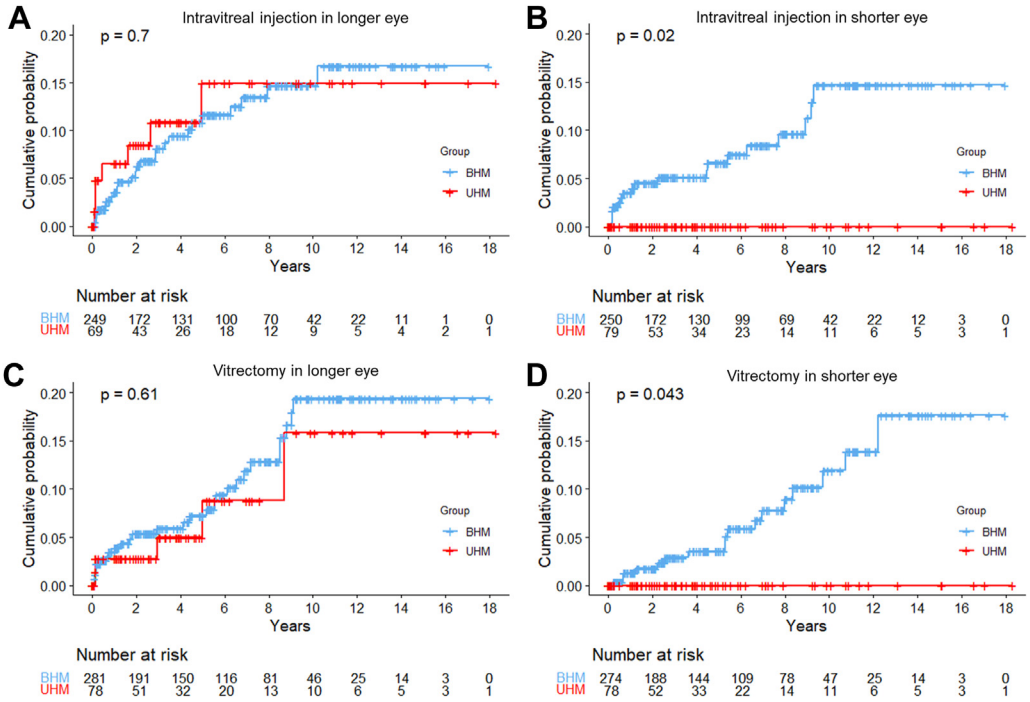


Figure 2. Cumulative probability of first myopia-related treatment in treatment-naïve patients. The first treatments for MNM (**A, B**) and MTM (**C, D**) occurred consistently over time in longer eyes, with no significant difference between the 2 groups (**A, C**). Additionally, in certain cases, treatment became necessary over time in the shorter eyes of the BHM group. However, in the UHM group, no cases required treatment for MTM or MNM over time (**B, D**). BHM = bilateral high myopia; MNM = myopic neovascular maculopathy; MTM = myopic tractional maculopathy; UHM = unilateral high myopia.

0.071 ± 0.006 mm/year for the shorter eye, both $P < 0.01$). When comparing the axial elongation rate between the 2 groups, there was no significant difference in the longer eye ($P = 0.09$); however, a significant difference was observed in the shorter eye ($P < 0.01$).

Discussion

In this study, we conducted a comparative analysis of the clinical characteristics of patients with UHM, with an average age of 57 years, against those with BHM. Previous

Table 3. Generalized Estimating Equations for Factors Affecting Axial Length

Parameters	Longer Eye		Shorter Eye	
	Coefficient ± SE	*P* Value	Coefficient ± SE	*P* Value
Intercept	30.639 ± 0.158		29.768 ± 0.165	
Sex		0.10		0.18
Female	Ref		Ref	
Male	0.651 ± 0.393		0.480 ± 0.357	
Cataract surgery during the follow-up period		<0.01		0.03
No	Ref		Ref	
Yes	−0.182 ± 0.038		−0.156 ± 0.070	
Vitrectomy during the follow-up period		0.28		0.28
No	Ref		Ref	
Yes	0.048 ± 0.044		0.121 ± 0.113	
Group		<0.01		<0.01
BHM	Ref		Ref	
UHM	−1.178 ± 0.271		−5.441 ± 0.209	
Time (BHM)	0.066 ± 0.006	<0.01	0.071 ± 0.006	<0.01
Time (UHM)	0.049 ± 0.010	<0.01	0.014 ± 0.008	0.12
Time X Group		0.09		<0.01
BHM	Ref		Ref	
UHM	−0.017 ± 0.010		−0.057 ± 0.006	

BHM = bilateral high myopia; SE = standard error; UHM = unilateral high myopia.

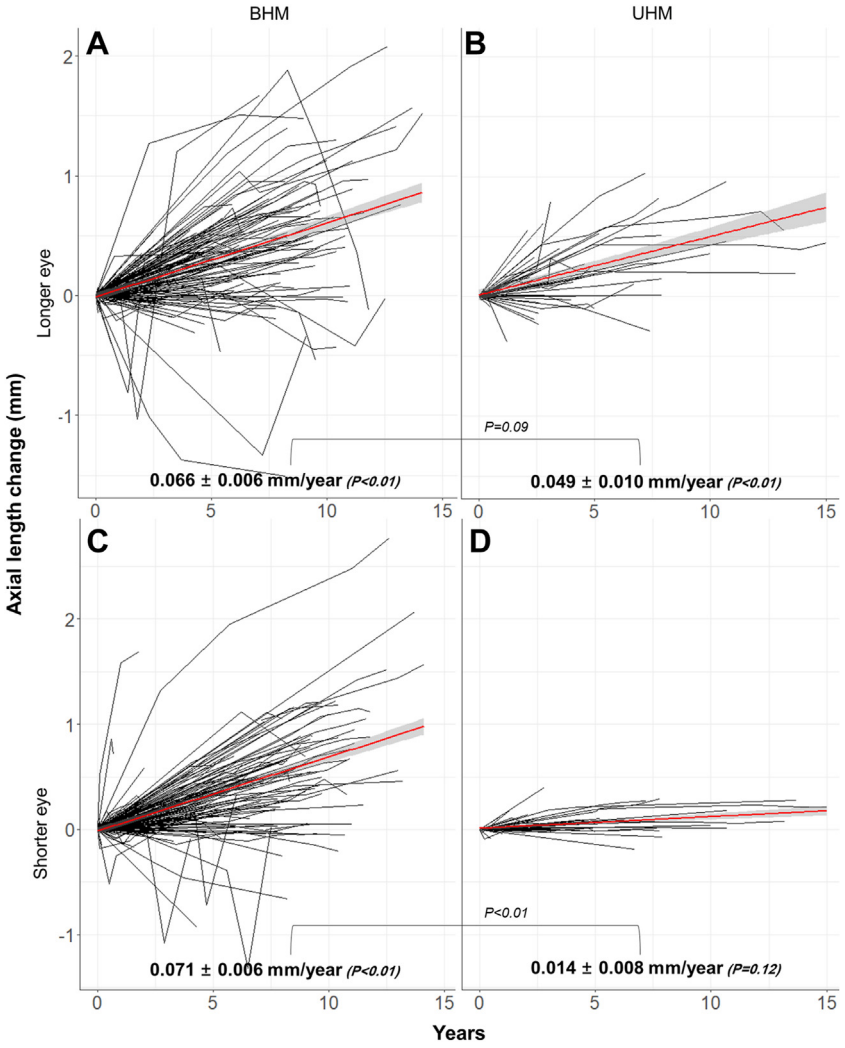


Figure 3. The spaghetti plot of the axial length changes over time. Both eyes in the BHM group and the longer eye in the UHM group showed significant axial length elongation over time (**A, B, C**). However, the shorter eye in the UHM group did not show significant axial length elongation over time (**D**). The rate of axial length elongation was not significantly different between the longer eyes of the 2 groups ($P = 0.09$), but there was a significant difference between the shorter eyes ($P < 0.01$). BHM = bilateral high myopia; UHM = unilateral high myopia.

studies on unilateral or monocular high myopia have predominantly focused on pediatric patients.[17,18] However, cases of asymmetric high myopia may have been temporarily classified as unilateral during the axial elongation process, which typically occurs until the mean age of 16 years in myopic children.[19] Therefore, we aimed to identify the clinical features of patients exhibiting UHM, even in adulthood, when the AL has stabilized.

Despite the longer eye of the UHM group having a shorter AL than the longer eye of the BHM group at the initial examination, the UHM group exhibited worse best-corrected visual acuity. This discrepancy was presumed to be due to the influence of anisometropic amblyopia, as there were no statistically significant differences in the ophthalmic treatment history, myopic maculopathy grade, or OCT parameters at the initial examination between the longer eyes of the 2 groups. This suggests that the interocular differences observed in these patients may have occurred at younger ages. Similar to previous studies on myopia, both groups included a higher proportion of females, consistent with existing research on myopia.[20] The proportion of females was slightly higher in the UHM group; however, there are no comparable studies focusing on this observation, and large-scale studies on anisometropia have reported no significant gender differences,[21] suggesting the need for further research.

Significant elongation of AL over time was observed in all highly myopic eyes, including both eyes in the BHM group and longer eyes in the UHM group. However, there was no significant increase in AL over time in the shorter eyes in the UHM group. The average rate of AL increase in all highly myopic eyes was approximately 0.05 to 0.07 mm per year, consistent with the axial elongation rates reported in previous studies on adults with and without high myopia.[20,22,23] However, such differences in the axial elongation pattern between the high myopic eye and the

nonhigh myopic eye of patients with UHM have not been reported. Collectively, it can be concluded that the shorter eyes in the UHM group represent a distinct natural progression, differing significantly from the less myopic eyes commonly observed in high myopia.

Notably, several studies have reported that anisometropia occurs in humans when unilateral image degradation occurs, such as blepharoptosis,[24] media opacities such as corneal scarring,[25] cataracts,[26] or vitreous hemorrhage.[27] Other hypotheses, including interocular differences in the cornea, intraocular pressure, convergence and accommodation, higher-order aberrations, and ocular dominance, are under investigation, although the evidence supporting these is limited.[28] Importantly, among the patients in the UHM group, only 2 had unilateral image degradation. One patient underwent corneal transplantation for keratoconus in the longer eye at the age of 30 years, and the other had corneal opacity with a history of childhood ocular trauma. Therefore, while previously reported unilateral image degradation may have caused unilateral myopia, it was not considered to be the primary mechanism in our cohort of patients.

There have been several reports suggesting a genetic influence in cases of severe anisometropia or asymmetric fundus findings. In monozygotic twins or siblings, severe anisometropia and asymmetric fundus findings have been reported on symmetrical or mirror images (involvement of different eyes in twins or siblings).[29–33] Feng et al[34] reported a Chinese family with an autosomal recessive pattern with 5.0 D myopic anisometropia. Weiss analyzed 48 children with UHM and reported 3 female patients with a strong family history.[17] However, these reports are based on small-scale studies, and further research or larger studies are needed.

In the wild, most animals exhibit a specific form of symmetry, with over 99% of the species having bilateral symmetry.[35] However, how 2 symmetrical cells on either side determine their equal positions and whether they are located on the left or right of the midplane is a fascinating phenomenon. Over the last 2 decades, numerous studies have been conducted to determine how animals achieve symmetry. One promising hypothesis suggests the presence of active mechanisms that protect symmetry from the impact of asymmetric influences.[36] For instance, Holt–Oram syndrome (OMIM: 142900) manifests as a dominant disorder primarily affecting the upper limbs in humans, often presenting with defects in the left arm. Holt–Oram syndrome is triggered by the haploinsufficiency of the forelimb factor, *TBX5*, which appears to play a role in maintaining limb symmetry in animal models.[37] While any asymmetry of other body parts was not found in our patients with UHM and the genetic factors involved in ocular symmetry remain unknown, considering recent studies on symmetry in other body parts, we speculate that similar genetic factors may also be involved in maintaining symmetry in the eyes.

This study had some limitations. First, its retrospective design led to variations in the total period and frequency of follow-up visits and examinations. Therefore, we were compelled to utilize methodologies such as GEE or survival analysis to analyze longitudinal change in quantitative variables, and there were limitations in analyzing the long-term clinical characteristics of qualitative variables, such as myopic maculopathy grade. Second, as this was a hospital-based cohort, there may have been differences in patient demographics and an inherent possibility of selection bias. Lastly, because our cohort includes patients who underwent CNV treatment or vitrectomy before their first visit to our center, the incidence proportion of MNM or MTM in survival analysis excluding such patients might underestimate their risks. However, given that a similar proportion of previous treatments was observed across all eyes except for those with shorter eyes in the UHM group. The impact of this limitation on the distinctive feature of UHM is considered to be minimal.

Regardless of the above limitations, this study focused on patients with UHM with an average interocular AL difference of 4.7 mm and a refractive error difference of 7.3 D. Considering the rarity of these cases based on previous epidemiological research on anisometropia,[21] we believe these cases are exceptionally uncommon. Particularly, the clinical profiles and natural progression of approximately 80 patients exhibiting such characteristics were examined over an average period of 4.8 years, adding significant value to this study.

In conclusion, UHM in adults exhibited unique disease features that distinguish it from typical high myopia. An apparent nonorganic visual impairment, suggestive of anisometropic amblyopia related to longstanding interocular asymmetry, was observed in the longer eyes in the UHM group despite the shorter AL than the longer eyes in the BHM group. The majority of shorter eyes in patients with UHM did not exhibit any characteristics of pathological myopia. There was no need for treatment for MNM or MTM during the follow-up period, and no axial elongation was observed over time in the shorter eyes of UHM. Further research on the underlying causal mechanisms, such as the genetic factors contributing to this extreme asymmetry, is thus necessary.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author.

Footnotes and Disclosures

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All authors have completed and submitted the ICMJE disclosures form.

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HUMAN SUBJECTS: Human subjects were included in this study. The study adhered to the tenets of the Declaration of Helsinki and was approved by the Institutional Review Board of Seoul National University Bundang Hospital (IRB approval number: B-2203-744-108). Owing to the nature of this study, written informed consent was waived.

No animal subjects were used in this study.

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Abbreviations and Acronyms:
AL = axial length; **BHM** = bilateral high myopia; **CNV** = choroidal neovascularization; **GEE** = generalized estimating equation; **MNM** = myopic neovascular maculopathy; **MTM** = myopic tractional maculopathy; **UHM** = unilateral high myopia.

Keywords:
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